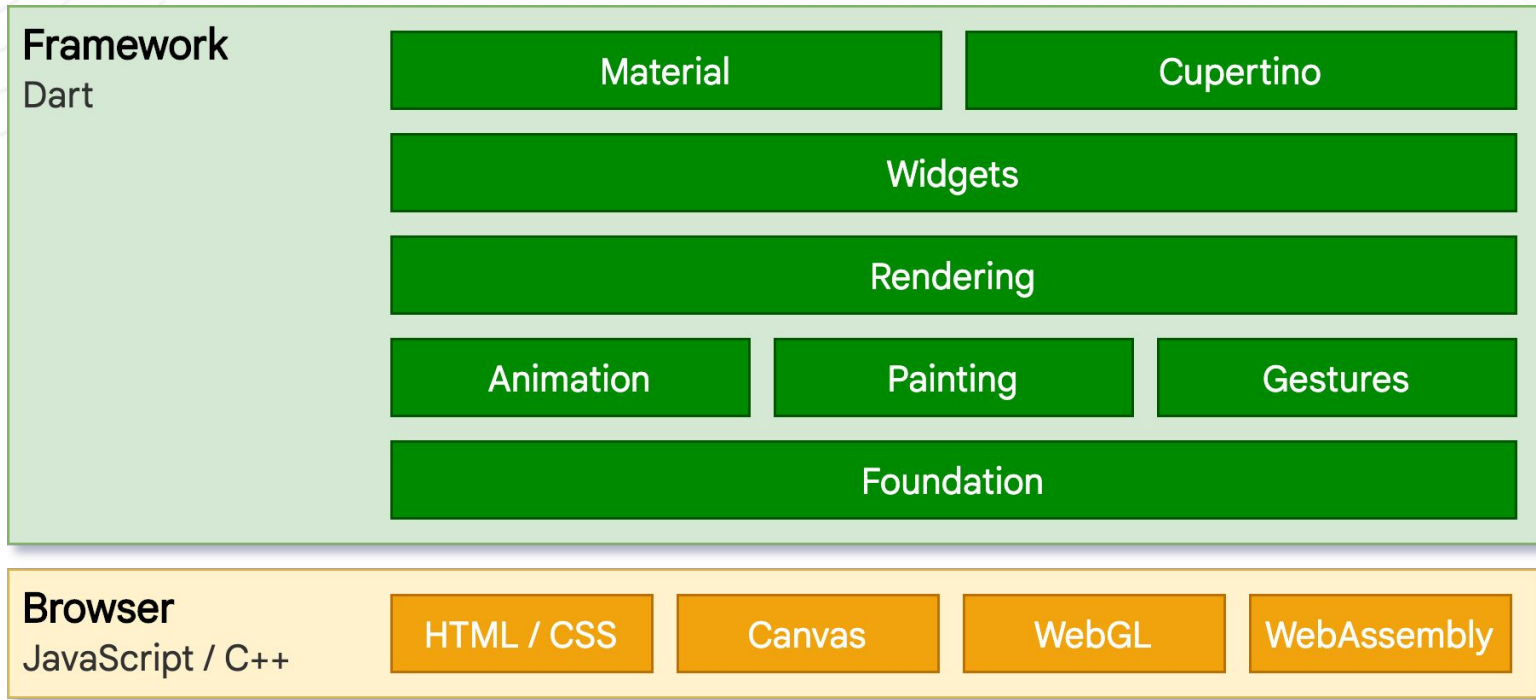


Flutter Web & Desktop

State of web and desktop in 2024

Flutter Web

How it works





Development toolchain
Fast incremental compilation
Stateful hot reload



Production toolchain
Fastest native output
Smallest runtime



ARM32



Risc-V



ARM64



x86_64



Wasm



JavaScript

Dart Native

Dart Web

How it works

Renderers

Flutter has two renderers (`canvaskit` and `skwasm`) that re-implement the Flutter engine to run the browser. The renderer converts UI primitives (stored as `Scene` objects) into pixels.

canvaskit

The `canvaskit` renderer is compatible with all modern browsers, and is the renderer that is used in the *default* build mode.

It includes a copy of Skia compiled to WebAssembly, which adds about 1.5MB in download size.

skwasm

The `skwasm` renderer is a more compact version of Skia that is compiled to WebAssembly and supports rendering on a separate thread.

This renderer must be used with the *WebAssembly* build mode, which compiles the Dart code to WebAssembly.

Use cases

1. Add web support for existing Flutter mobile apps
2. SPA (Single Page Application)
3. PWA (but you might need some [workarounds](#))

Things that can't be used on web:

1. dart:io (you can use conditional imports or packages like [universal io](#))
2. dart:isolate
3. dart:ffi

Things that work only on the web

New interop libraries

```
package:web
```

```
dart:js_interop  
dart:js_interop_unsafe
```

Previous libraries

```
dart:html  
dart:indexed_db  
dart:svg  
dart:web_audio  
dart:web_gl
```

```
package:js  
dart:js  
dart:js_util
```

See this [lint rule](#)

Conditional imports

```
export 'src/hw_none.dart' // Stub implementation
if (dart.library.io) 'src/hw_io.dart' // dart:io implementation
if (dart.library.js_interop) 'src/hw_web.dart'; // package:web implementation
```

```
import 'dart:io';

void alarm([String? text]) {
  stderr.writeln(text ?? message);
}

String get message => 'Hello World from the VM!';
```

```
import 'package:hw_mp/hw_mp.dart';

void main() {
  print(message);
}
```

```
void alarm([String? text]) => throw UnsupportedError('hw_none alarm');

String get message => throw UnsupportedError('hw_none message');
```

LayoutBuilder and other responsive widgets

```
body: LayoutBuilder(  
  builder: (BuildContext context, BoxConstraints constraints) {  
    if (constraints.maxWidth > 600) {  
      return _buildWideContainers();  
    } else {  
      return _buildNormalContainer();  
    }  
  },  
,
```

```
  child: FractionallySizedBox(  
    widthFactor: 0.5,  
    heightFactor: 0.5,  
    alignment: FractionalOffset.center,  
    child: DecoratedBox(  
      decoration: BoxDecoration(  
        border: Border.all(  
          color: Colors.blue,  
          width: 4,  
        ),  
      ),  
    ),  
  ),  
,
```

Image support on web

CanvasKit renderer

- Uses WebGL to render images
- Subject to CORS policy
- Not optimized by the browser

See all limitations and workarounds

<https://docs.flutter.dev/platform-integration/web/web-images>

Caveats

Hot reload doesn't work on web

<https://github.com/flutter/flutter/issues/53041>

Startup time (resolved)

<https://github.com/flutter/flutter/issues/76009>

main.dart.js too large and code splitting

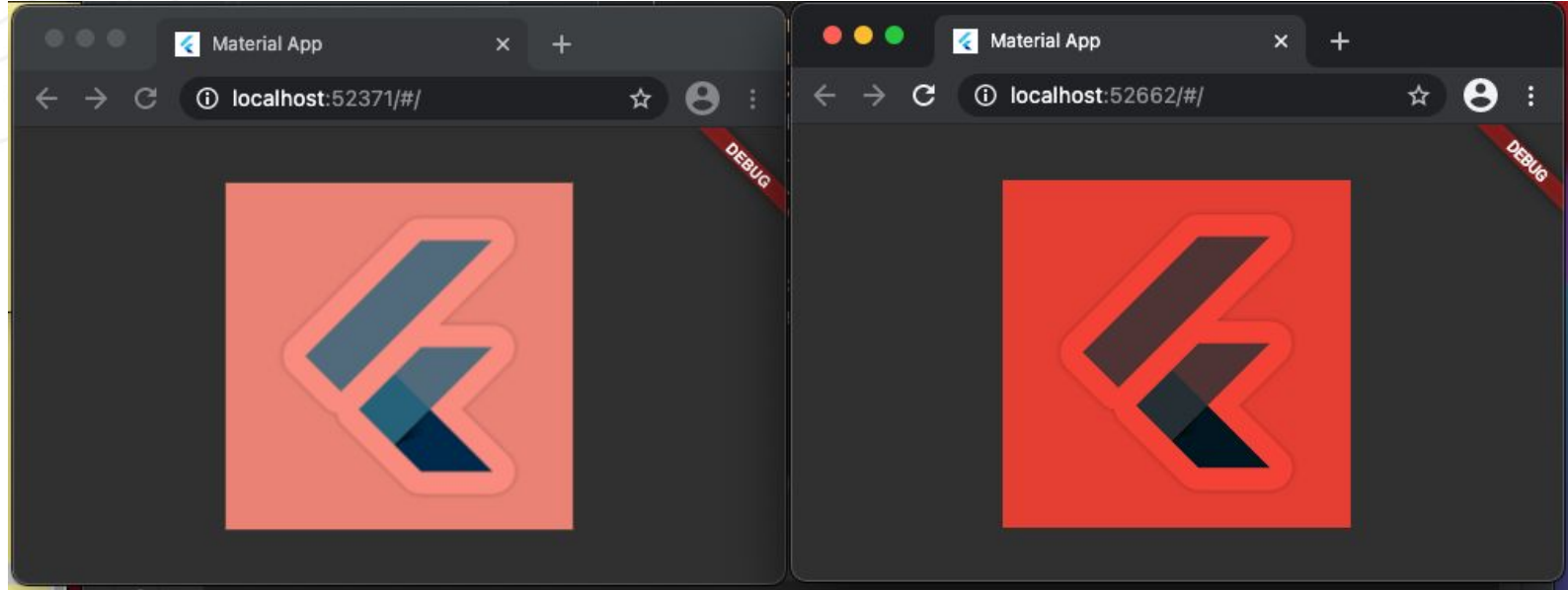
<https://github.com/flutter/flutter/issues/46589>

Deferred loading (web “only”)

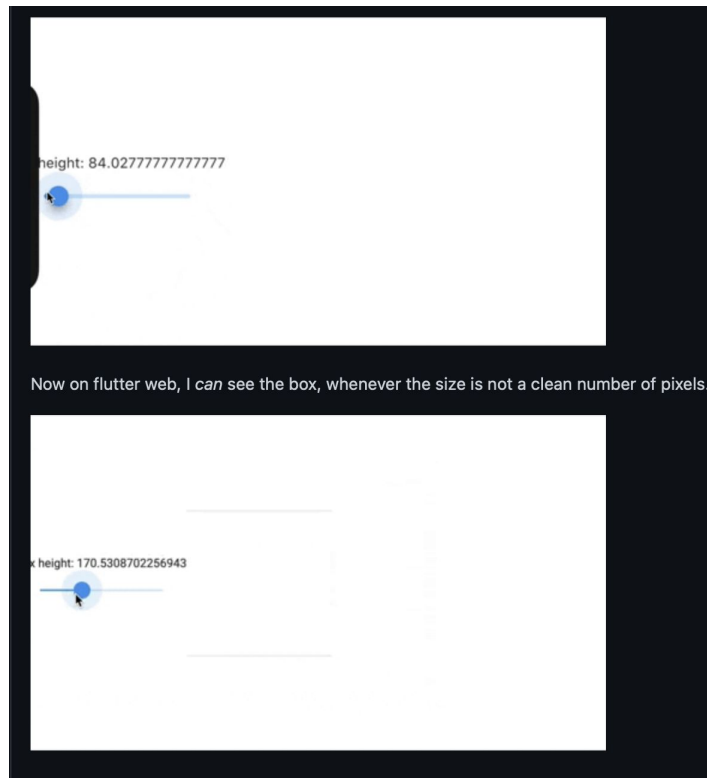
```
import 'package:greetings/hello.dart' deferred as hello;
```

```
Future<void> greet() async {  
  await hello.loadLibrary();  
  hello.printGreeting();  
}
```

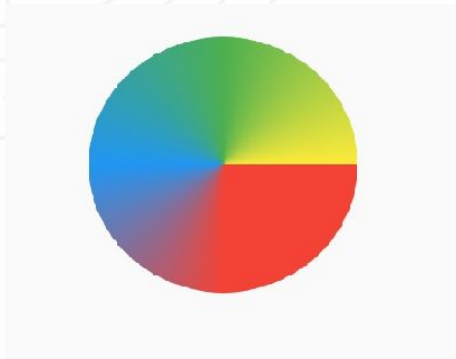

CanvasKit vs HTML differences - color blending



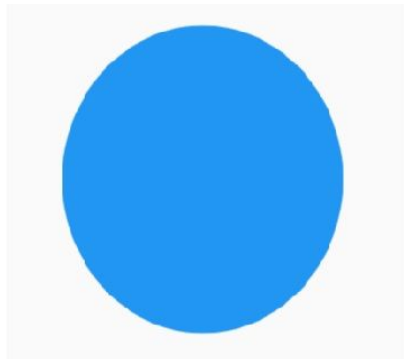
CanvasKit vs HTML differences - anti-aliasing



CanvasKit vs HTML differences - anti-aliasing



Canvas kit



html

376 Another exception was thrown: Unsupported operation: ImageShader not implemented for web platform. [js_primitives.dart:30](#)

Another exception was thrown: Unsupported operation: ImageShader not implemented for web platform. [js_primitives.dart:30](#)

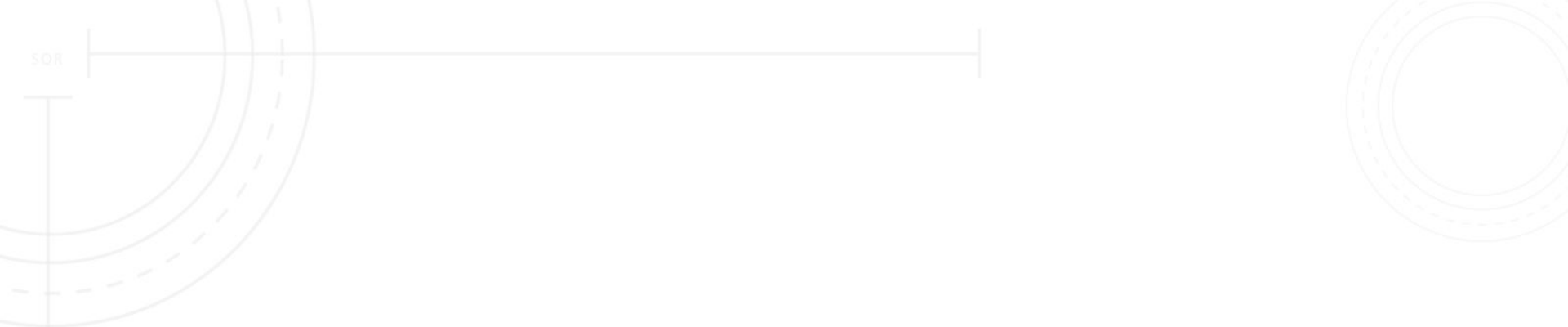


canvas kit

<https://github.com/flutter/flutter/issues/84984>

SEO issues

<https://github.com/flutter/flutter/issues/46789>



Caching issues

<https://github.com/flutter/flutter/issues/63500>

Element & multi-view embedding

<https://docs.flutter.dev/platform-integration/web/embedding-flutter-web>

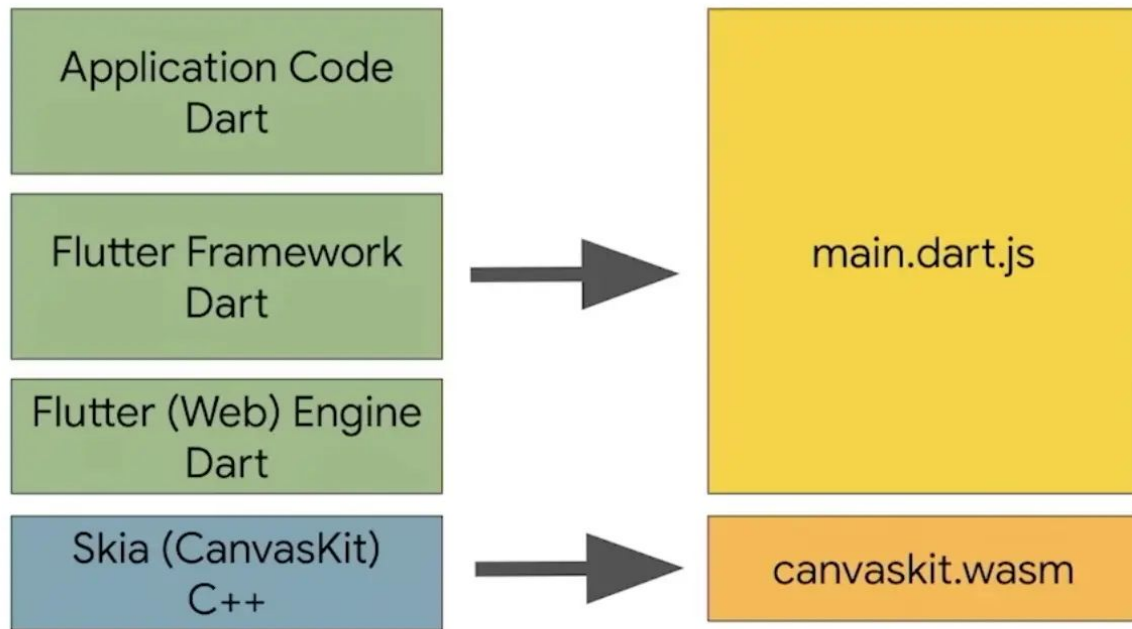
Multi-view element embedding not supported yet

<https://github.com/flutter/flutter/issues/118481>

What's next for Flutter on the Web?

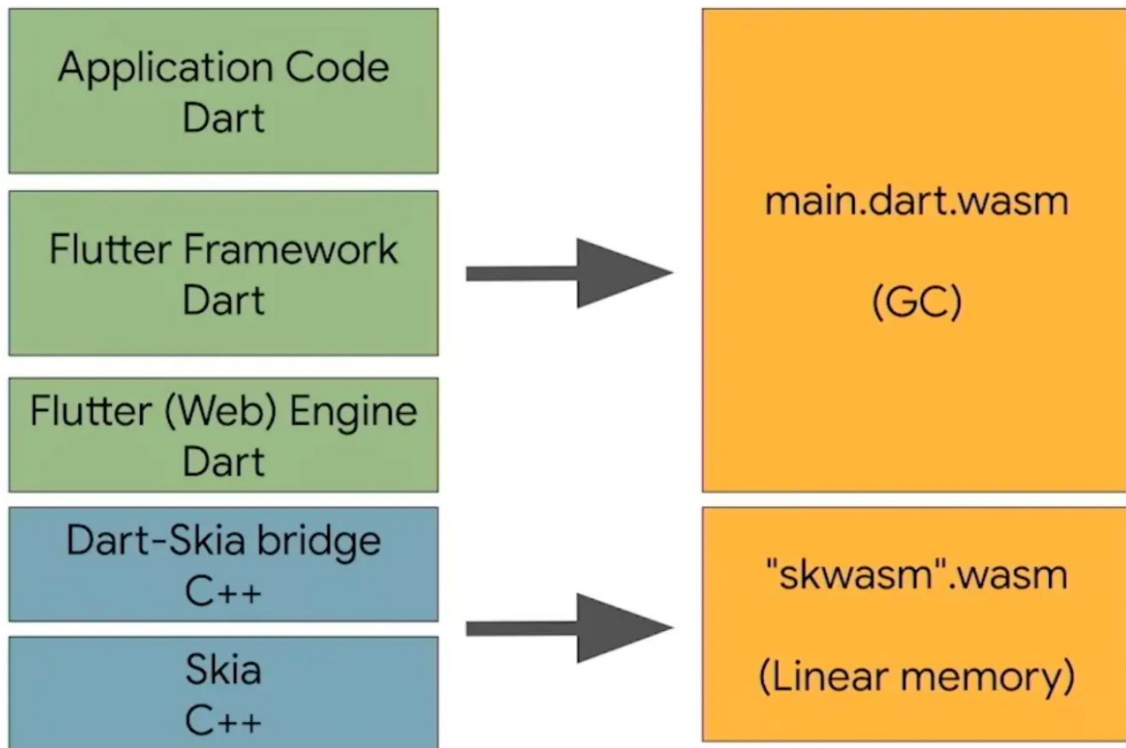
Starting with WebAssembly

State as of
March 2023

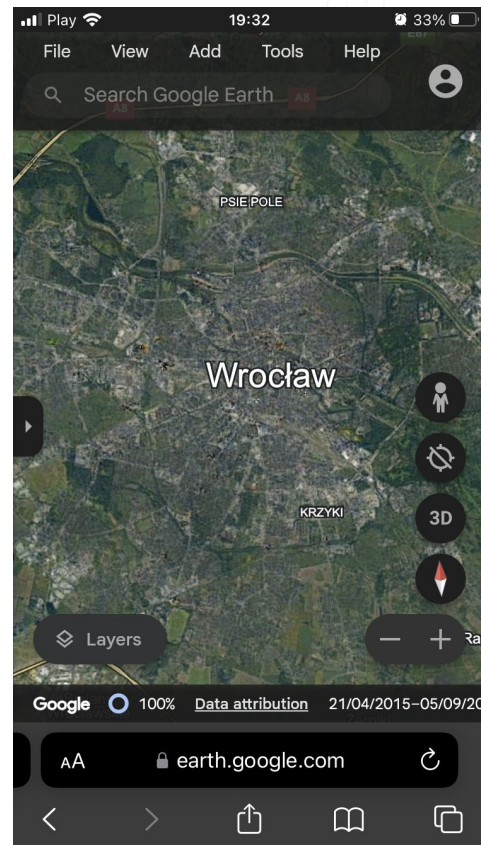
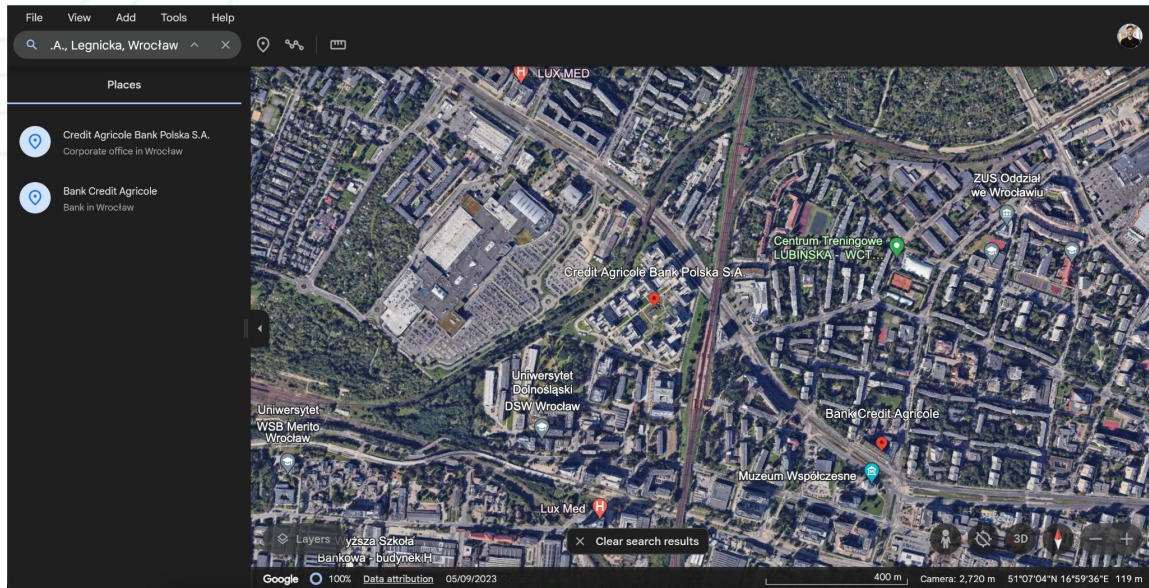


All-in on WebAssembly

Our wasm **plan** as of
March 2023



Google Earth - earth.google.com



Uses Dart WebAssembly support

Flutter Desktop

Supported deployment platforms

As of Flutter 3.27.0, Flutter supports deploying apps on the following combinations of hardware architectures and operating system versions. These combinations are called *platforms*.

Flutter categorizes platforms as follows:

- **Supported:** The platforms and versions that the Flutter team supports.
- **CI-tested:** The Flutter team tests these platforms on every commit.
- **Unsupported:** The Flutter team doesn't test or support these platforms.

Target platform	Target architectures	Supported versions	CI-tested versions	Unsupported versions
Android SDK	x64, Arm32, Arm64	21 to 34	21 to 34	20 and earlier
iOS	Arm64	12 to 18	17	11 and earlier
macOS	x64, Arm64	Mojave (10.14) to Sequoia (15)	Ventura (13), Sonoma (14)	High Sierra (10.13) and earlier
Windows	x64, Arm64	10, 11	10	8 and earlier
Debian (Linux)	x64, Arm64	10, 11, 12	11, 12	9 and earlier
Ubuntu (Linux)	x64, Arm64	20.04 LTS to 24.04 LTS	20.04 LTS, 22.04 LTS	23.10 and earlier non-LTS
Chrome (Web)	JavaScript, WebAssembly	Latest 2	119, 125	95 and earlier
Firefox (Web)	JavaScript	Latest 2	132	98 and earlier
Safari (Web)	JavaScript	15.6 and newer	15.6	15.5 and earlier
Edge (Web)	JavaScript, WebAssembly	Latest 2	119, 125	95 and earlier

Enable desktop support

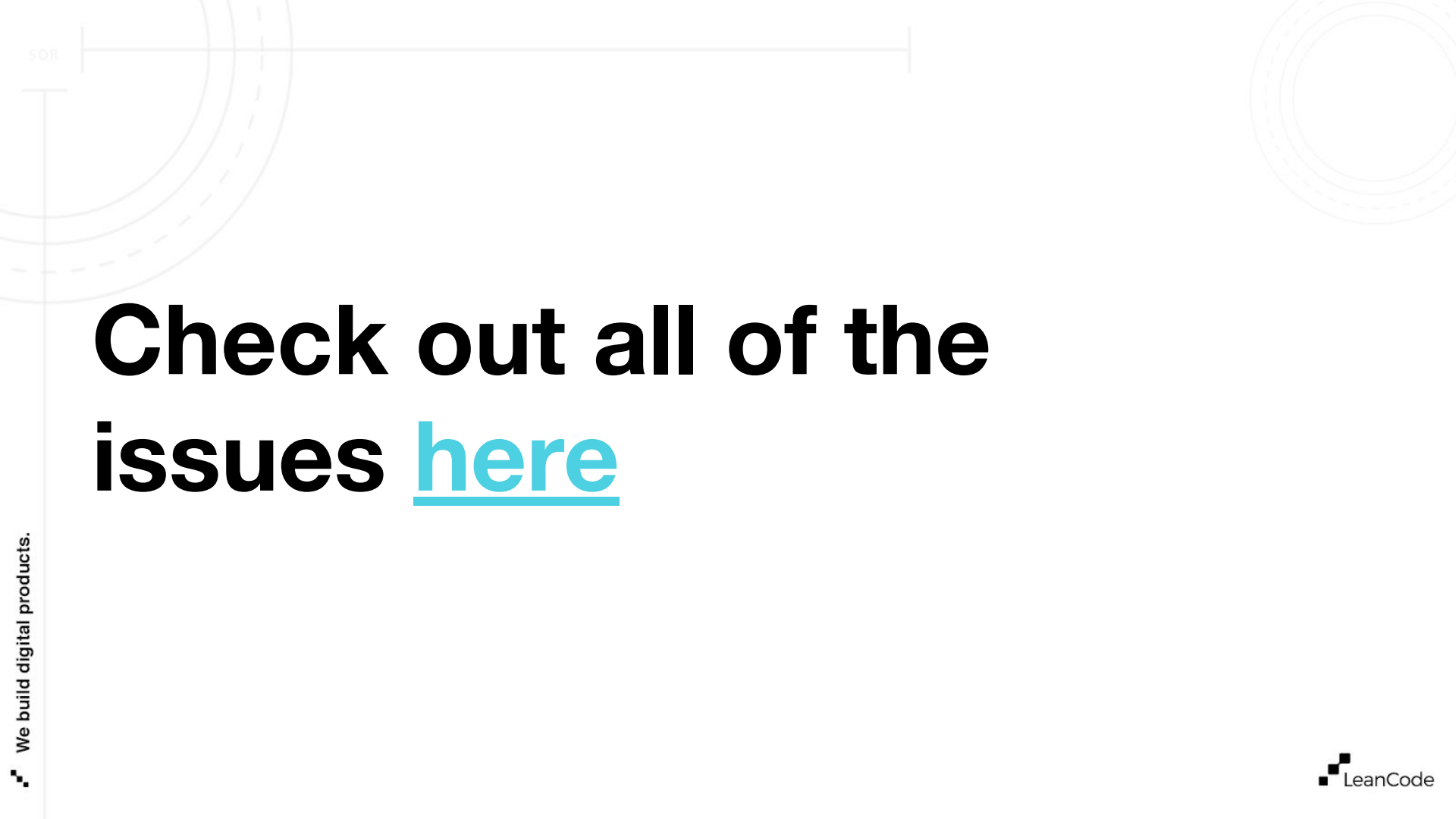
```
$ flutter config --enable-windows-desktop # for the Windows runner  
$ flutter config --enable-macos-desktop  # for the macOS runner  
$ flutter config --enable-linux-desktop   # for the Linux runner
```

No PlatformView implementation

<https://github.com/flutter/flutter/issues/41722>
<https://github.com/flutter/flutter/issues/31713>

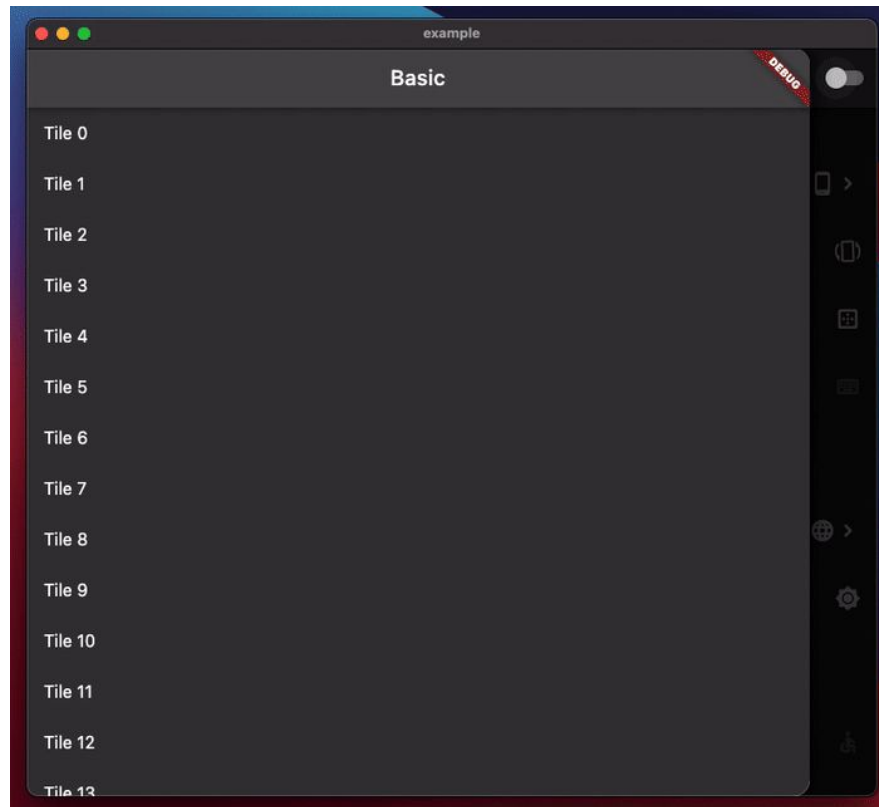
Multiple windows support

<https://github.com/flutter/flutter/issues/30701>



**Check out all of the
issues [here](#)**

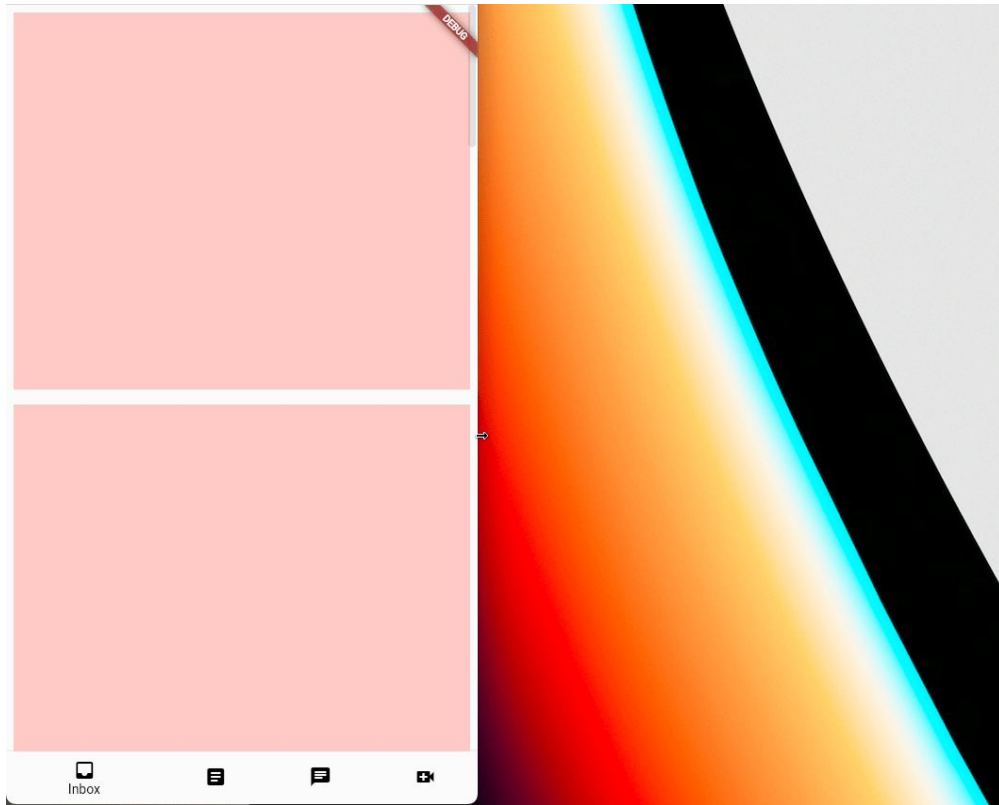
device preview



Adaptive layouts

- `AspectRatio`
- `CustomSingleChildLayout`
- `CustomMultiChildLayout`
- `FittedBox`
- `FractionallySizedBox`
- `LayoutBuilder`
- `MediaQuery`
- `MediaQueryData`
- `OrientationBuilder`

flutter_adaptive_scaffold



Custom embedders

<https://github.com/flutter/flutter/wiki/Custom-Flutter-Engine-Embedders>



WE
ROMAN
JUST
CODES

Flutter Multiplatform

Sources

[Docs / flutter.dev](https://docs.flutter.dev)

github.com/flutter/flutter

Kevin Moore @ Wasm I/O 2023

youtube.com/@romanjustcodes/